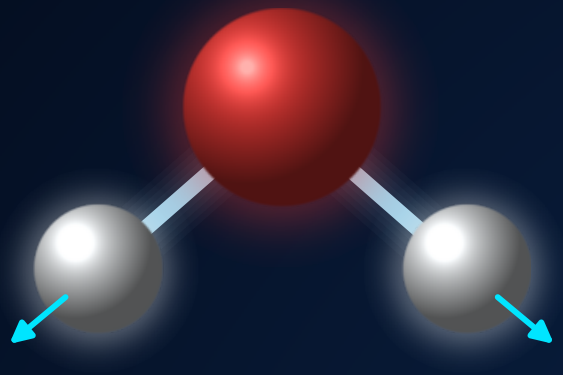
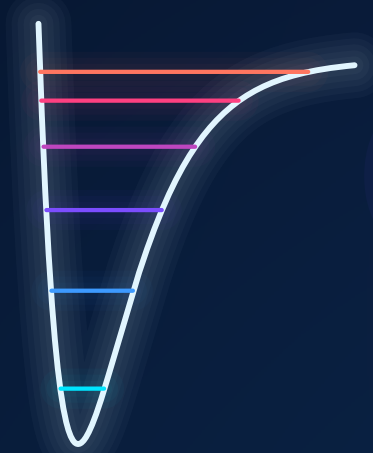
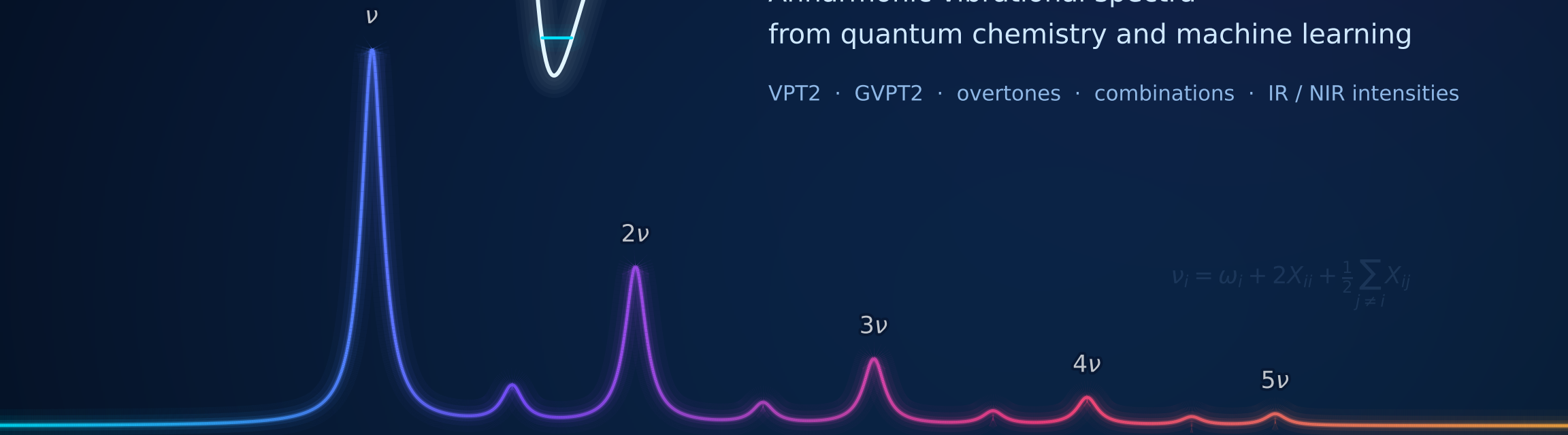




OVERTONE

Anharmonic vibrational spectra
from quantum chemistry and machine learning

VPT2 · GVPT2 · overtones · combinations · IR / NIR intensities



$$A_{0f} \propto \tilde{\nu}_{0f} |\langle 0|\boldsymbol{\mu}|f\rangle|^2$$

$$G(\mathbf{v}) = \sum_r \omega_r v_r + \sum_r X_{rr}(v_r^2 + v_r) + \sum_{r<s} X_{rs} v_r v_s$$

$$X_{ij} = \frac{\phi_{iiii}}{16} - \sum_k \frac{\phi_{iik}^2(8\omega_i^2 - 3\omega_k^2)}{16\,\omega_k(4\omega_i^2 - \omega_k^2)}$$

$$\nu_i = \omega_i + 2X_{ii} + \frac{1}{2}\sum_{j\neq i} X_{ij}$$

$$V(q) = \frac{1}{2}\sum_i \omega_i q_i^2 + \frac{1}{6}\sum \phi_{ijk} q_i q_j q_k + \frac{1}{24}\sum \phi_{ijkl} q_i q_j q_k q_l$$

$$\frac{\partial E}{\partial q} \approx \frac{-E_{+2} + 8E_{+1} - 8E_{-1} + E_{-2}}{12\,\delta}$$